## Importing libraries and SIFT init.

```
import cv2 as cv
import matplotlib.pyplot as plt
import numpy as np
# Initialize SIFT detector
sift = cv.SIFT_create()

print("OpenCV imported and SIFT detector initialized.")
```

```
OpenCV imported and SIFT detector initialized.
```

## Load Image and Video

```
# Load the image
img = cv.imread('/content/sample_data/cv_cover.jpg')

# Check if image was loaded successfully
if img is None:
    print("Error: Could not load image from /content/sample_data/cv_cover.jpg")
else:
    print("Image loaded successfully.")

# Load the video
cap = cv.VideoCapture('/content/sample_data/book.mov')

# Check if video was loaded successfully
if not cap.isOpened():
    print("Error: Could not open video from /content/sample_data/book.mov")
else:
    print("Video loaded successfully.")

    # Read the first frame of the video
    ret, frame = cap.read()
    if not ret:
        print("Error: Could not read first frame from video.")
    else:
        print("First video frame read successfully.")

    # Release the video capture object
    cap.release()


# Display the loaded image and frame (optional, for verification)
# cv2_imshow(img) # Uncomment if you have cv2_imshow available or convert to matplotlib
# cv2_imshow(frame) # Uncomment if you have cv2_imshow available or convert to matplotlib
```

```
Image loaded successfully.
Video loaded successfully.
First video frame read successfully.
```

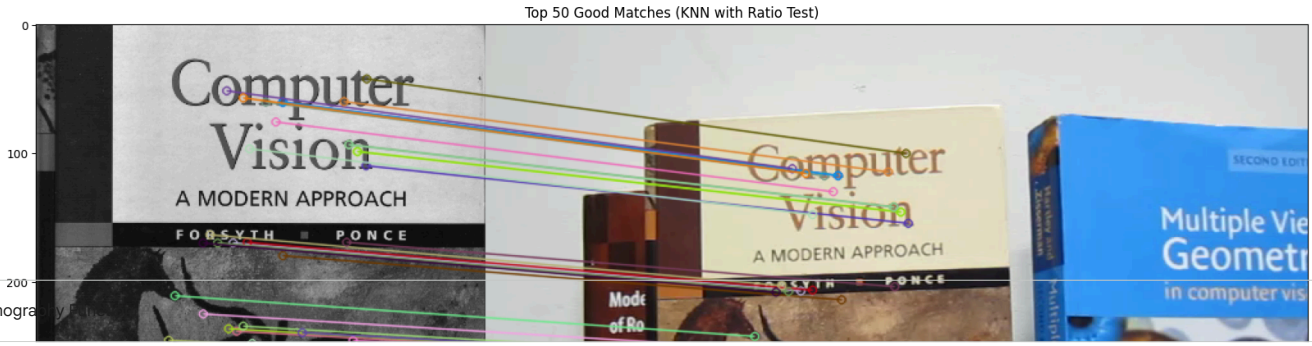
## Convert to Grayscale and Find Keypoints

```
# Convert image to grayscale
if img is not None:
    gray_img = cv.cvtColor(img, cv.COLOR_BGR2GRAY)
    # Find keypoints and descriptors for the image
    kp_img, des_img = sift.detectAndCompute(gray_img, None)
    print(f"Found {len(kp_img)} keypoints in the image.")
else:
    kp_img, des_img = None, None

# Convert video frame to grayscale
if 'frame' in locals() and frame is not None:
    gray_frame = cv.cvtColor(frame, cv.COLOR_BGR2GRAY)
    # Find keypoints and descriptors for the video frame
    kp_frame, des_frame = sift.detectAndCompute(gray_frame, None)
    print(f"Found {len(kp_frame)} keypoints in the video frame.")
else:
    kp_frame, des_frame = None, None

print("Keypoints and descriptors found for image and video frame (if loaded successfully).")
```

```
Found 1205 keypoints in the image.
Found 1842 keypoints in the video frame.
Keypoints and descriptors found for image and video frame (if loaded successfully).
```

## Using Brute force matcher

```
bfm = cv.BFMatcher_create(cv.NORM_L2) # Remove Cross-Check

# Find k=2 best matches for each descriptor
matches = bfm.knnMatch(des_img, des_frame, k=2)

# Apply ratio test to filter good matches
good_matches = []
for m, n in matches:
    if m.distance < 0.7 * n.distance:
        good_matches.append(m)

# Sort good_matches by distance and take the top N
good_matches = sorted(good_matches, key=lambda x: x.distance)
nMatches = 50

# Draw only the top 'nMatches' good matches
img_matches = cv.drawMatches(
    img, kp_img,
    frame, kp_frame,
    good_matches[:nMatches],
    None,
    flags=cv.DrawMatchesFlags_NOT_DRAW_SINGLE_POINTS
)

plt.figure(figsize=(20, 10))
# Convert BGR image to RGB for matplotlib display
plt.imshow(cv.cvtColor(img_matches, cv.COLOR_BGR2RGB))
plt.title(f'Top {min(nMatches, len(good_matches))} Good Matches (KNN with Ratio Test)')
plt.show()
```




## Homography Fn

```
def findHomography(pts_src, pts_dst):
  if len(pts_src) < 4 or len(pts_dst) < 4:
      raise ValueError("At least 4 corresponding points are required to compute a homography.")
  if len(pts_src) != len(pts_dst):
      raise ValueError("Number of source and destination points must be equal.")

  n = pts_src.shape[0]
  A = np.zeros((2 * n, 8))
  b = np.zeros((2 * n, 1))

  for i in range(n):
      x, y = pts_src[i]
      xp, yp = pts_dst[i]

      # Construct rows for the A matrix based on the equation:
      # x' = (h11*x + h12*y + h13) / (h31*x + h32*y + 1)
      # => h11*x + h12*y + h13 - h31*x*x' - h32*y*x' = x'
      A[2 * i, 0] = x
      A[2 * i, 1] = y
      A[2 * i, 2] = 1
      A[2 * i, 6] = -x * xp
      A[2 * i, 7] = -y * xp
      b[2 * i] = xp

      # Equation for y' = (h21*x + h22*y + h23) / (h31*x + h32*y + 1)
      # => h21*x + h22*y + h23 - h31*x*y' - h32*y*y' = y'
      A[2 * i + 1, 3] = x
      A[2 * i + 1, 4] = y
      A[2 * i + 1, 5] = 1
      A[2 * i + 1, 6] = -x * yp
      A[2 * i + 1, 7] = -y * yp
      b[2 * i + 1] = yp

  # Solve the linear system Ah_vec = b_vec for h_vec (8 unknowns)
  h_vec, _, _, _ = np.linalg.lstsq(A, b, rcond=None)

  # Reshape the 8-vector into a 3x3 Homography matrix, setting H[2,2] = 1
  H = np.zeros((3, 3))
  H[0, 0] = h_vec[0, 0]
  H[0, 1] = h_vec[1, 0]
  H[0, 2] = h_vec[2, 0]
  H[1, 0] = h_vec[3, 0]
  H[1, 1] = h_vec[4, 0]
  H[1, 2] = h_vec[5, 0]
  H[2, 0] = h_vec[6, 0]
  H[2, 1] = h_vec[7, 0]
  H[2, 2] = 1.0 # As per the problem statement

  return H
```

```
if good_matches:
    # Extract location of good matches
    # Source points (from img)
    pts_src = np.float32([kp_img[m.queryIdx].pt for m in good_matches]).reshape(-1, 1, 2)
    # Destination points (from frame)
    pts_dst = np.float32([kp_frame[m.trainIdx].pt for m in good_matches]).reshape(-1, 1, 2)

    # Calculate homography using the custom function
    H_custom = findHomography(pts_src[:, 0], pts_dst[:, 0])
    print("Custom Homography Matrix H_custom:")
    print(H_custom)

    # Test the homography with a few points
    # Let's pick the first 5 good matches for testing
    test_pts_src = pts_src[:5]
    test_pts_dst_actual = pts_dst[:5]

    print("\nTesting Homography with 5 points:")
    for i in range(len(test_pts_src)):
        src_pt = test_pts_src[i][0]
        actual_dst_pt = test_pts_dst_actual[i][0]

        # Convert source point to homogeneous coordinates (x, y, 1)
        src_pt_homogeneous = np.array([src_pt[0], src_pt[1], 1]).reshape(3, 1)

        # Transform point using homography
        transformed_pt_homogeneous = np.dot(H_custom, src_pt_homogeneous)

        # Convert back to Cartesian coordinates (x', y')
        transformed_pt = (transformed_pt_homogeneous[0:2] / transformed_pt_homogeneous[2]).flatten()

        print(f"  Source: {src_pt}")
        print(f"  Transformed: {transformed_pt}")
        print(f"  Actual Destination: {actual_dst_pt}")
        print(f"  Difference: {np.linalg.norm(transformed_pt - actual_dst_pt):.4f}\n")
else:
    print("No good matches found to calculate homography.")
```

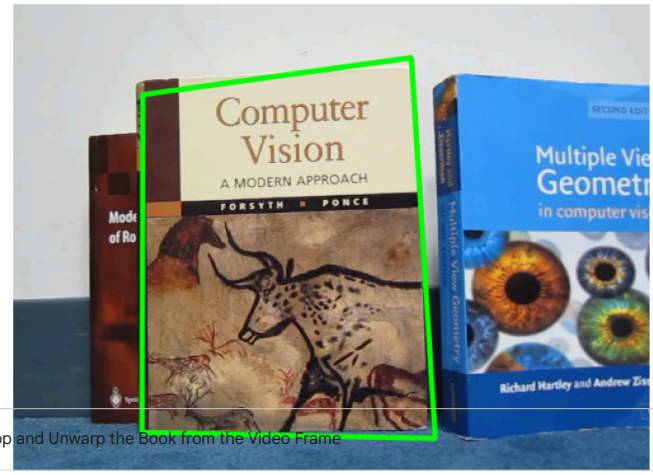
```
Custom Homography Matrix H_custom:
[[ 6.45525585e-01 -2.57253845e-02  1.30798465e+02]
 [-1.23968102e-01  6.97703908e-01  9.17752634e+01]
 [-3.19093244e-04 -1.73776858e-04  1.00000000e+00]]

Testing Homography with 5 points:
  Source: [129.84949 225.25644]
  Transformed: [227.12630852 253.24658416]
  Actual Destination: [226.94148 253.17487]
  Difference: 0.1983

  Source: [169.9911  254.79167]
  Transformed: [259.54811463 275.62553677]
  Actual Destination: [260.78482 277.00305]
  Difference: 1.8512
```

```
Source: [149.65019 236.72469]
Transformed: [242.90324006 261.64464941]
Actual Destination: [243.73103 262.51007]
Difference: 1.1976

Source: [191.79314   60.753185]
Transformed: [272.60431722 118.92016812]
Actual Destination: [273.90848  117.780266]
Difference: 1.7321

Source: [107.89213 211.09807]
Transformed: [209.94450634 242.96140443]
Actual Destination: [209.10095 242.74455]
Difference: 0.8710
```

## Calculate and Draw Book Coordinates in Video Frame

```
# Get the dimensions of the image (cover)
h, w = img.shape[:2]

# Define the four corners of the image
pts = np.float32([[0, 0], [0, h - 1], [w - 1, h - 1], [w - 1, 0]]).reshape(-1, 1, 2)

# Apply the homography to transform the corners to the video frame
dst = cv.perspectiveTransform(pts, H_custom)

# Draw the transformed corners on the frame
# Create a copy of the frame to draw on, so the original frame remains unchanged
frame_with_book = frame.copy()
frame_with_book = cv.polylines(frame_with_book, [np.int32(dst)], True, (0, 255, 0), 3, cv.LINE_AA)

# Display the result
plt.figure(figsize=(10, 8))
plt.imshow(cv.cvtColor(frame_with_book, cv.COLOR_BGR2RGB))
plt.title('Book Detected in Video Frame')
plt.axis('off')
plt.show()
```

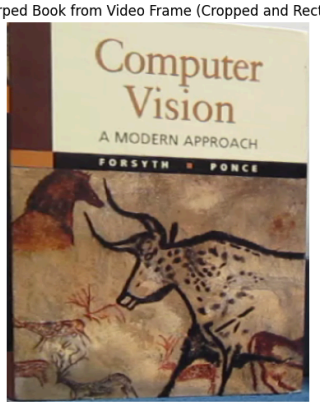
Book Detected in Video Frame




## Crop and Unwarp the Book from the Video Frame

```
# Calculate the inverse homography matrix
H_custom_inverse = np.linalg.inv(H_custom)

# Get the dimensions of the original image (cover)
h_img, w_img = img.shape[:2]

# Apply the inverse homography to the video frame to unwarp the book
# The output size (dsize) will be the dimensions of the original image
warped_book = cv.warpPerspective(frame, H_custom_inverse, (w_img, h_img))

# Display the unwarped book
plt.figure(figsize=(8, 6))
plt.imshow(cv.cvtColor(warped_book, cv.COLOR_BGR2RGB))
plt.title('Unwarped Book from Video Frame (Cropped and Rectified)')
plt.axis('off')
plt.show()
```

Unwarped Book from Video Frame (Cropped and Rectified)



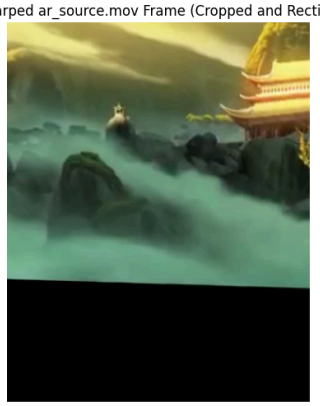

## Crop and Unwarp 'ar_source.mov' for Overlay

```
# Load the ar_source video
cap_ar = cv.VideoCapture('/content/sample_data/ar_source.mov')
```

```
# Check if video was loaded successfully
if not cap_ar.isOpened():
    print("Error: Could not open video from /content/sample_data/ar_source.mov")
    ar_frame = None
else:
    print("ar_source.mov video loaded successfully.")

    # Read the first frame of the video
    ret_ar, ar_frame = cap_ar.read()
    if not ret_ar:
        print("Error: Could not read first frame from ar_source.mov.")
        ar_frame = None
    else:
        print("First ar_source.mov frame read successfully.")

    # Release the video capture object
    cap_ar.release()

if ar_frame is not None:
    # Apply the inverse homography to the ar_frame
    # The output size (dsize) will be the dimensions of the original image (cover)
    warped_ar = cv.warpPerspective(ar_frame, H_custom_inverse, (w_img, h_img))

    # Display the unwarped ar_source frame
    plt.figure(figsize=(8, 6))
    plt.imshow(cv.cvtColor(warped_ar, cv.COLOR_BGR2RGB))
    plt.title('Unwarped ar_source.mov Frame (Cropped and Rectified)')
    plt.axis('off')
    plt.show()
else:
    print("Cannot proceed with ar_source.mov as the frame was not loaded.")
```

```
ar_source.mov video loaded successfully.
First ar_source.mov frame read successfully.
```

Unwarped ar_source.mov Frame (Cropped and Rectified)



## Overlay 'ar_source' onto 'book.mov' frame for AR Effect

```
# Ensure 'frame' and 'ar_frame' are loaded
if frame is None or ar_frame is None:
    print("Error: 'frame' or 'ar_frame' is not loaded. Cannot proceed with overlay.")
else:
    # Get the dimensions of the video frame
    h_frame, w_frame = frame.shape[:2]

    # 1. Create a mask for the region where the book is in the 'frame'
    # The 'dst' variable holds the transformed corners of the book in the 'frame'
    mask = np.zeros((h_frame, w_frame), dtype=np.uint8)
    cv.fillConvexPoly(mask, np.int32(dst), 255)

    # 2. Invert the mask to get the background of the frame
    mask_inv = cv.bitwise_not(mask)

    # 3. Use the inverse mask to black out the book area in the original frame
    frame_bg = cv.bitwise_and(frame, frame, mask=mask_inv)

    # 4. Warp the 'ar_frame' to the perspective of the book in the 'frame'
    # The output size should be the size of the 'frame' so it can be directly overlaid
    ar_warped_to_frame = cv.warpPerspective(ar_frame, H_custom, (w_frame, h_frame))

    # 5. Use the original mask to get the foreground (warped ar_source) only where the book is
    ar_fg = cv.bitwise_and(ar_warped_to_frame, ar_warped_to_frame, mask=mask)

    # 6. Combine the background of the original frame with the warped ar_source foreground
    ar_result = cv.add(frame_bg, ar_fg)

    # Display the result
    plt.figure(figsize=(12, 8))
    plt.imshow(cv.cvtColor(ar_result, cv.COLOR_BGR2RGB))
    plt.title('Augmented Reality Effect: ar_source overlaid on book_frame')
    plt.axis('off')
    plt.show()
```

Augmented Reality Effect: ar_source overlaid on book_frame

## Create Final AR

```
# Re-open video capture objects for full video processing
cap_book = cv.VideoCapture('/content/sample_data/book.mov')
cap_ar = cv.VideoCapture('/content/sample_data/ar_source.mov')

if not cap_book.isOpened() or not cap_ar.isOpened():
    print("Error: Could not open one or both video files.")
else:
    # Get video properties
    frame_width = int(cap_book.get(cv.CAP_PROP_FRAME_WIDTH))
    frame_height = int(cap_book.get(cv.CAP_PROP_FRAME_HEIGHT))
    fps = cap_book.get(cv.CAP_PROP_FPS)

    # Define the codec and create VideoWriter object
    # Use XVID for wider compatibility or MP4V/DIVX
    fourcc = cv.VideoWriter_fourcc(*'XVID')
    out = cv.VideoWriter('ar_output.avi', fourcc, fps, (frame_width, frame_height))

    print(f"Processing video (FPS: {fps}, Resolution: {frame_width}x{frame_height})...")

    while True:
        ret_book, frame_book = cap_book.read()
        ret_ar, frame_ar = cap_ar.read()

        if not ret_book or not ret_ar:
            break # Break the loop if no more frames are available from either video

        # Convert current book frame to grayscale for SIFT
        gray_frame_book = cv.cvtColor(frame_book, cv.COLOR_BGR2GRAY)

        # Find keypoints and descriptors for the current book frame
        kp_frame_book, des_frame_book = sift.detectAndCompute(gray_frame_book, None)

        ar_result_frame = frame_book.copy() # Default to original frame_book

        if des_frame_book is not None and len(kp_frame_book) > 0:
            # Match descriptors between the cover image and the current book frame
            matches_video = bfm.knnMatch(des_img, des_frame_book, k=2)

            good_matches_video = []
            for m, n in matches_video:
                if m.distance < 0.7 * n.distance:
                    good_matches_video.append(m)

            if len(good_matches_video) > 10: # Ensure enough matches for stable homography
                # Extract location of good matches for homography calculation
                pts_src_video = np.float32([kp_img[m.queryIdx].pt for m in good_matches_video]).reshape(-1, 1, 2)
                pts_dst_video = np.float32([kp_frame_book[m.trainIdx].pt for m in good_matches_video]).reshape(-1, 1, 2)

                # Calculate homography for the current frame
                H_current_frame, mask_homography = cv.findHomography(pts_src_video, pts_dst_video, cv.RANSAC, 5.0)

                if H_current_frame is not None:
                    # 1. Create a mask for the region where the book is in the 'frame_book'
                    h_book_img, w_book_img = img.shape[:2]
                    pts_corners = np.float32([[0, 0], [0, h_book_img - 1], [w_book_img - 1, h_book_img - 1], [w_book_img - 1, 0]]).reshape(-1, 1, 2)
                    dst_corners_current = cv.perspectiveTransform(pts_corners, H_current_frame)

                    mask_overlay = np.zeros((frame_height, frame_width), dtype=np.uint8)
                    cv.fillConvexPoly(mask_overlay, np.int32(dst_corners_current), 255)

                    # 2. Invert the mask for the background
                    mask_inv_overlay = cv.bitwise_not(mask_overlay)

                    # 3. Use the inverse mask to black out the book area in the original frame_book
                    frame_book_bg = cv.bitwise_and(frame_book, frame_book, mask=mask_inv_overlay)

                    # 4. Warp the current 'ar_frame' to the perspective of the book in 'frame_book'
                    ar_warped_to_current_frame = cv.warpPerspective(frame_ar, H_current_frame, (frame_width, frame_height))

                    # 5. Use the original mask to get the foreground (warped ar_source) only where the book is
                    ar_fg_overlay = cv.bitwise_and(ar_warped_to_current_frame, ar_warped_to_current_frame, mask=mask_overlay)

                    # 6. Combine the background of frame_book with the warped ar_source foreground
                    ar_result_frame = cv.add(frame_book_bg, ar_fg_overlay)
            else:
                print(f"Warning: Not enough good matches ({len(good_matches_video)}) for homography in current frame. Skipping AR overlay.")

        # Write the processed frame to the output video
        out.write(ar_result_frame)

    print("Video processing complete. Output saved to ar_output.avi")
    cap_book.release()
    cap_ar.release()
    out.release()
    cv.destroyAllWindows()
```

```
Processing video (FPS: 30.0, Resolution: 640x480)...
Video processing complete. Output saved to ar_output.avi
```

## Compare Original and AR Videos Side-by-Side

```
cap_original = cv.VideoCapture('/content/sample_data/book.mov')
cap_ar_output = cv.VideoCapture('/content/ar_output.avi')

if not cap_original.isOpened() or not cap_ar_output.isOpened():
    print("Error: Could not open one or both video files for comparison.")
else:
    # Get properties from the original video (assuming AR output has same dims and fps)
    frame_width_orig = int(cap_original.get(cv.CAP_PROP_FRAME_WIDTH))
    frame_height_orig = int(cap_original.get(cv.CAP_PROP_FRAME_HEIGHT))
    fps_orig = cap_original.get(cv.CAP_PROP_FPS)

    # Define the codec and create VideoWriter object for the combined video
    fourcc_combined = cv.VideoWriter_fourcc(*'XVID')
    out_combined = cv.VideoWriter('comparison_output.avi', fourcc_combined, fps_orig, (frame_width_orig * 2, frame_height_orig))

    print("Creating comparison video...")

    while True:
        ret_orig, frame_orig = cap_original.read()
        ret_ar, frame_ar = cap_ar_output.read()

        if not ret_orig or not ret_ar:
            break # Break the loop if no more frames are available from either video

        # Ensure both frames have the same dimensions if they somehow differ
        if frame_orig.shape != frame_ar.shape:
            frame_ar = cv.resize(frame_ar, (frame_width_orig, frame_height_orig))
```